Deriving the Dual

The general case 1/3

$$(P)\quad \left[\begin{array}{ll} \min: & c^T x \\ \text{st:} & Ax \le b \end{array}\right. \longleftrightarrow \min_x : \max_{p \le 0} : c^T x + p^T(b - Ax)$$

$$\geq \text{(weak duality)}$$

$$\max_{p \le 0} : \min_x : c^T x + p^T(b - Ax)$$

$$= \max_{p \le 0} : \min_x : \underbrace{(c^T - p^T A)}\, x + p^T b$$

$$(D)\quad = \begin{array}{ll} \max: & p^T b \\ \text{st:} & p^T A = c^T \\ & p \le 0 \end{array} \qquad (A^T p = c)$$

Strong Duality:
In fact we have equality throughout.

Deriving the Dual

The general case 2/3

$$\min: c^T x$$
$$\text{st}: Ax \le b$$
$$x \ge 0$$

$$\min_{x \le 0}: \max_{p \le 0}: c^T x + p^T(b - Ax)$$

$$\ge$$

$$\max_{p \le 0}: \min_{x \le 0}: (c^T - p^T A)x + p^T b$$

$$\max: p^T b$$
$$\text{st}: c^T - p^T A \le 0 \Leftrightarrow p^T A \ge c^T$$
$$p \le 0$$

Deriving the Dual

The general case 3/3

$$\min: c^T x$$

$$\text{st}: \quad A_1 x = b_1 \qquad p_1$$

$$A_2 x \geq b_2 \qquad p_2$$

$$x \leq 0$$

$$\min_{x \leq 0}: \max_{\substack{p_1 \\ p_2 \geq 0}}: c^T x + p_1^T (b_1 - A_1 x) + p_2^T (b_2 - A_2 x)$$

$$\geq$$

$$\max_{\substack{p_2 \geq 0 \\ p_1}}: \min_{x \leq 0}: \underbrace{(c^T - p_1^T A_1 - p_2^T A_2)}\, x + p_1^T b_1 + p_2^T b_2$$

$$\max: \quad p_1^T b_1 + p_2^T b_2$$

$$p_1^T A_1 + p_2^T A_2 \geq c$$

$$p_1 \text{ free}, \quad p_2 \geq 0$$

Deriving the Dual

An Example

$$\min: \; x_1 - x_2 + 0\cdot x_3 + 0\cdot x_4$$

$$\text{st:} \; 2x_1 + 3x_2 - x_3 + x_4 \le 0$$

$$3x_1 + x_2 + 4x_3 - 2x_4 \ge 3$$

$$-x_1 - x_2 + 2x_3 + x_4 = 6$$

$$\underline{x_1 \le 0}, \quad \underline{x_2, x_3 \ge 0}$$

Dual:

$$\max: \; 0p_1 + 3p_2 + 6p_3$$

$$\text{st:} \; 2p_1 + 3p_2 - p_3 \le 1$$

$$3p_1 + p_2 - p_3 \ge -1$$

$$p_1 \quad \cancel{3p_1}$$

$$p_2 \quad -p_1 + 4p_2 + 2p_3 \ge 0$$

$$p_3 \quad p_1 - 2p_2 + p_3 = 0$$

$$p_1 \le 0 \quad p_2 \ge 0 \quad p_3$$

$$\min: \max: \; x_1 - x_2 + p_1\big(0 - (2x_1 + 3x_2 - x_3 + x_4)\big) + p_2\big(3 - (3x_1 + x_2 + 4x_3 - 2x_4)\big) + p_3\big(6 - (-x_1 - x_2 + 2x_3 + x_4)\big)$$

$x_1 \le 0$, $x_2, x_3 \ge 0$; $p_1 \le 0$, $p_2 \ge 0$, p_3 free

$$\ge$$

$$\max: \min: \; (1 - 2p_1 - 3p_2 + p_3)x_1 + (-1 - 3p_1 - 1\cdot p_2 + 1p_3)x_2 + (0 + p_1 - 4p_2 - 2p_3)x_3 + (0 - p_1 - 2p_2 - p_3)x_4 + 0p_1 + 3p_2 + 6p_3$$

$p_1 \le 0$, $p_2 \ge 0$, p_3 free; $x_1 \le 0$, $x_2, x_3 \ge 0$

Example Two